DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING (A)
Osmania University, Hyderabad-07

CLASS TEST-I

Subject :Computer Organization and Microprocessors

Class: B.E (CSE)
Semester: III
Academic Year: 2025-2026

Max Marks: 30
Duration : 1 1/2
Date: 22-09-2025

		Marks	BT	CO
	PART A **Answer All Questions (5 x 2 = 10 M)**			
1. a)	Define an interrupt. What are the different types of interrupts?	2	2	1
b)	Define the following i) Micro operation ii) Microinstruction.	2	2	1
c)	Compare the hard wired control unit and micro programmed control unit.	2	4	2
d)	Write about the different phases in Instruction cycle.	2	2	1
e)	What are the functions of ALU?	2	2	2
	PART B **Answer Any Two Questions (2 x 10 = 20 M)**			
2.	a) Differentiate between fixed point and floating point representation.	5	4	1
	b) What is Instruction code? Explain direct and indirect address with an example.	5	2	1
3.	a) Draw the block diagram of an 8-bit ALU with a 4-bit status register and explain the purpose of each bit in the status register.	6	2	2
	b) List all the register reference instructions and explain about any four instructions.	4	2	1
4.	Illustrate the micro-programmed control unit (MCU) with a neat diagram and how instructions will be processed in MCU?	10	2	2